

Структура данных STL

queue — BFS

stack — вычисление выражений

deque

set < pg

pair

map

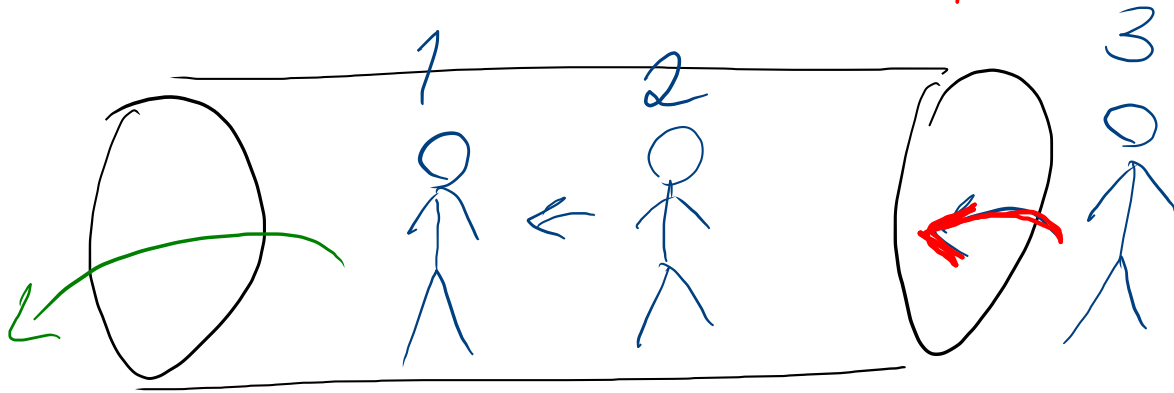
multiset

queue aka очередь aka

FIFO

back - последний $O(1)$

push



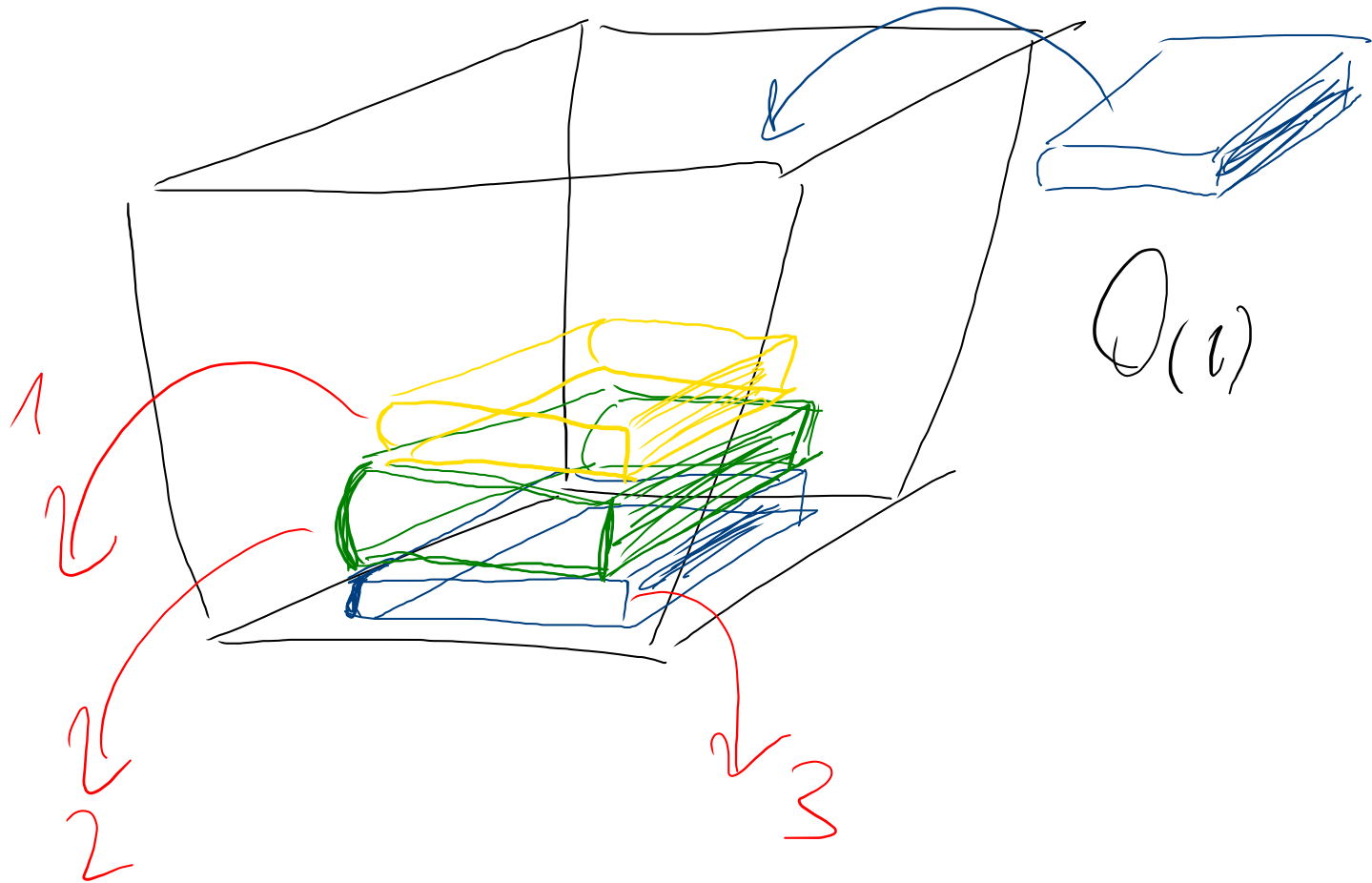
void

pop

front - первый в очереди

$O(1)$

stack aka CTOR aka **LIFO**



D3

$$7 \ 12 + * \rightarrow (1+2) * 7$$

$$\underline{7} \ \underline{1} \ \underline{2} + \underline{*} \ \underline{3} \ \underline{9} \ \underline{-} \ \underline{*}$$

stack = [~~7~~ ~~1~~ ~~2~~ ~~3~~ ~~21~~ ~~3~~ ~~9~~ ~~-6~~ -126]

Diagram illustrating the evaluation of the expression $7 \ 12 + * \rightarrow (1+2) * 7$ using a stack:

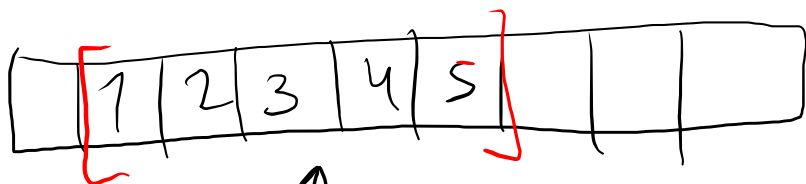
- The stack initially contains $7, 1, 2, 3$.
- The operation $+$ is applied to 1 and 2 , resulting in 3 . The stack becomes $[7, 3, 3]$.
- The operation $*$ is applied to 7 and 3 , resulting in 21 . The stack becomes $[21, 3]$.
- The operation $-$ is applied to 3 and 9 , resulting in -6 . The stack becomes $[21, -6]$.
- The operation $*$ is applied to 21 and -6 , resulting in -126 .

$$\text{ans} = -126 = (1+2) * 7 * (3-9)$$

deque

vs

list



macsub tuna

pop-back

back

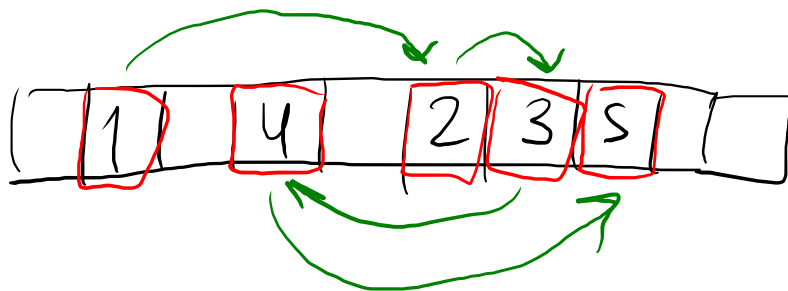
push-back

pop-front

front

push front

$O(1)$



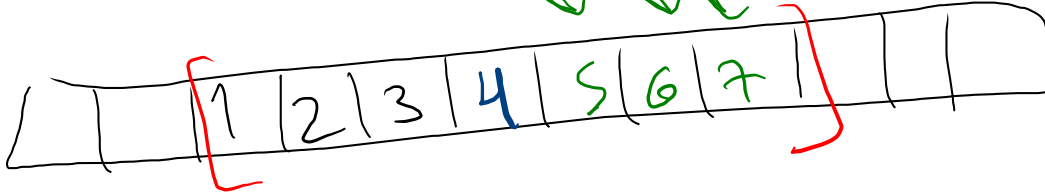
мөгөрөөг үз-зг ptr

4

V



$\sim O(n)$



set aka множество

хранит элементы в отсортированном
порядке без повторений
 n -много элементов

`set<string>`
 $O(\log_2 n \cdot m)$ $m = |S|$

insert, erase, find $O(\log_2 n)$

Р9

push, pop $O(\log_2 n)$

top $O(1)$

set

vs

unordered_set

отсортировано

НЕ отсортировано



find, erase, insert

~~begin~~

$O(\log_2 n)$

Red-Black tree

$O(1)$ амортизир.

Хеш-таблица

Python set